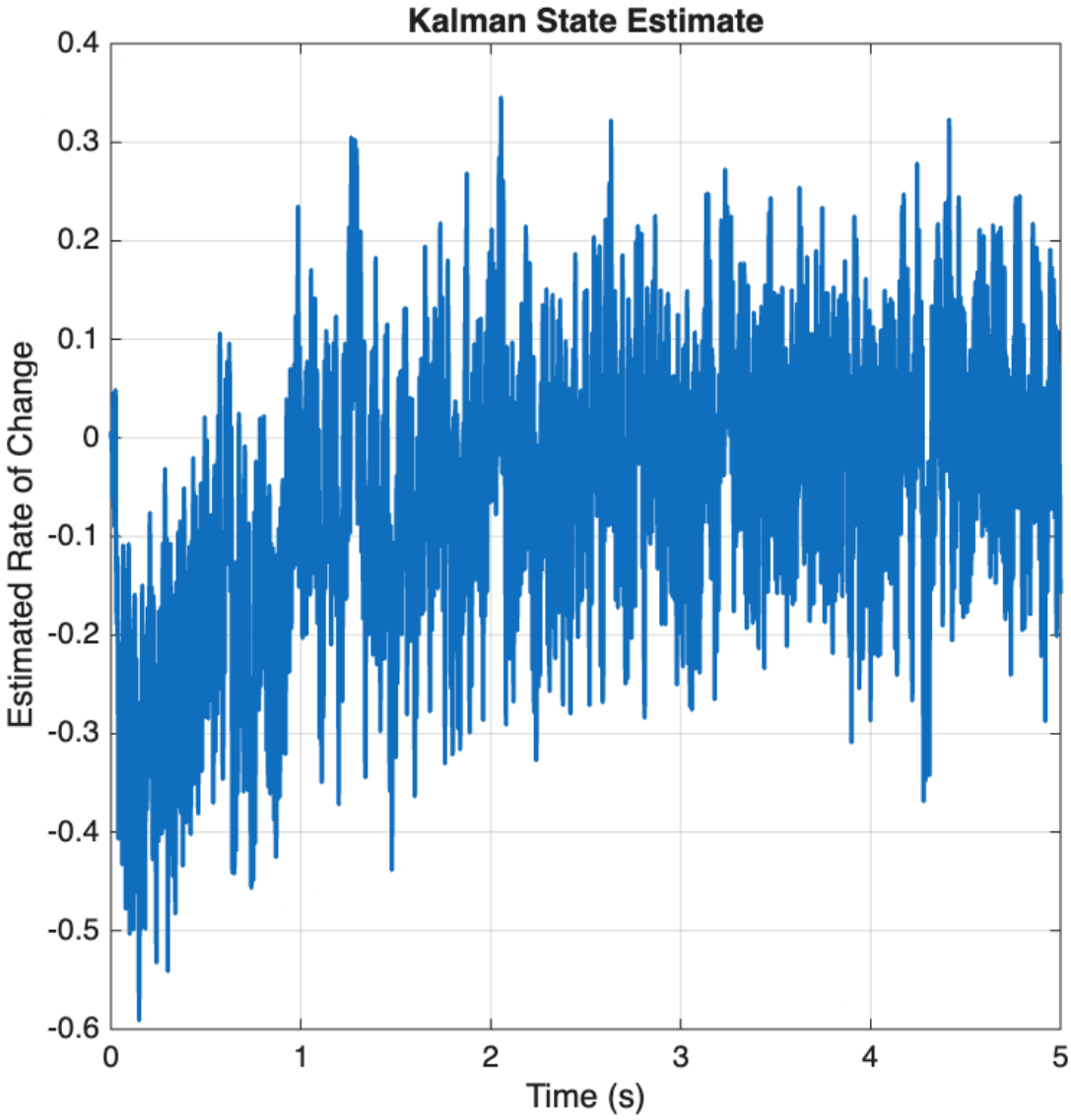
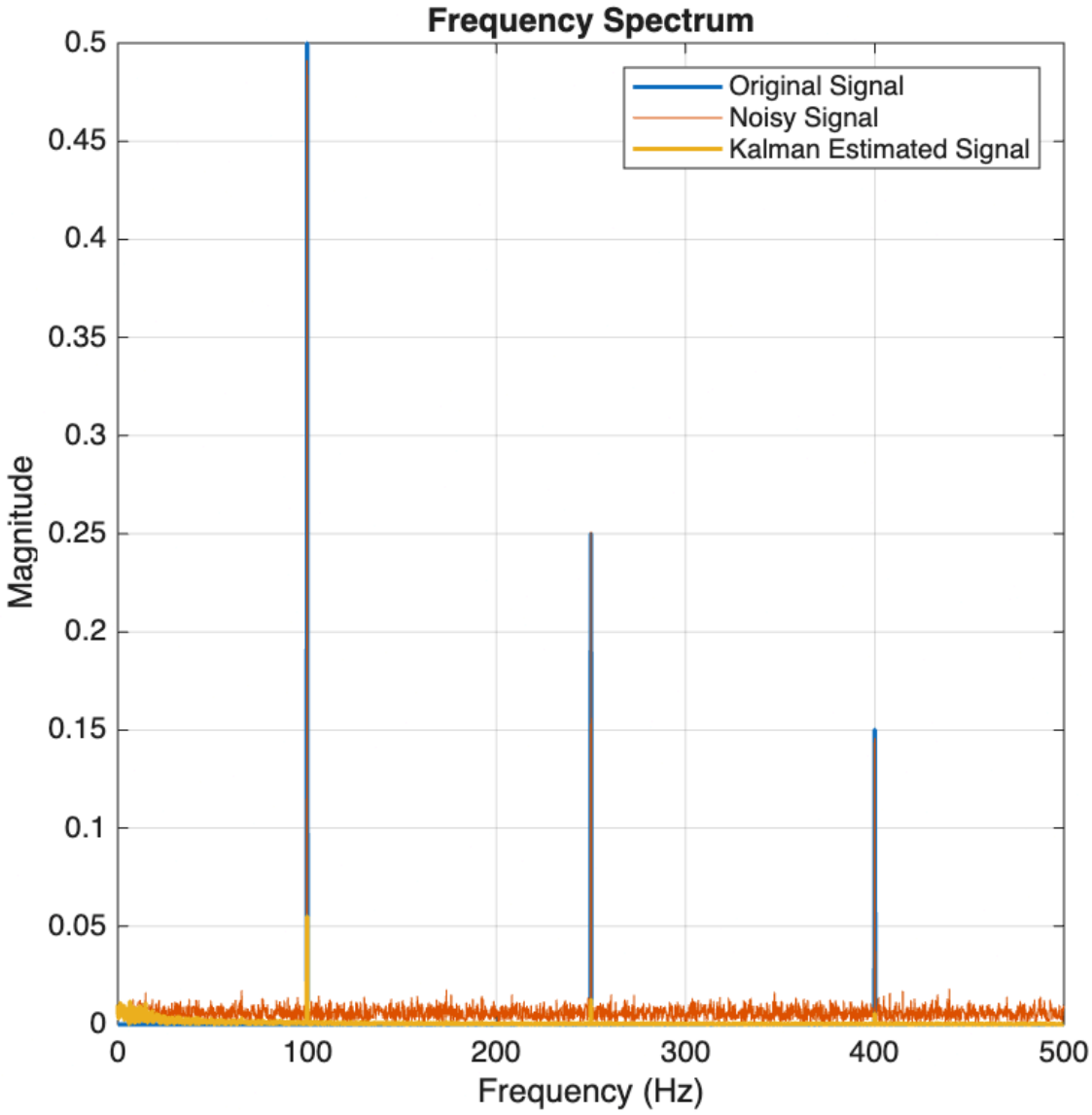
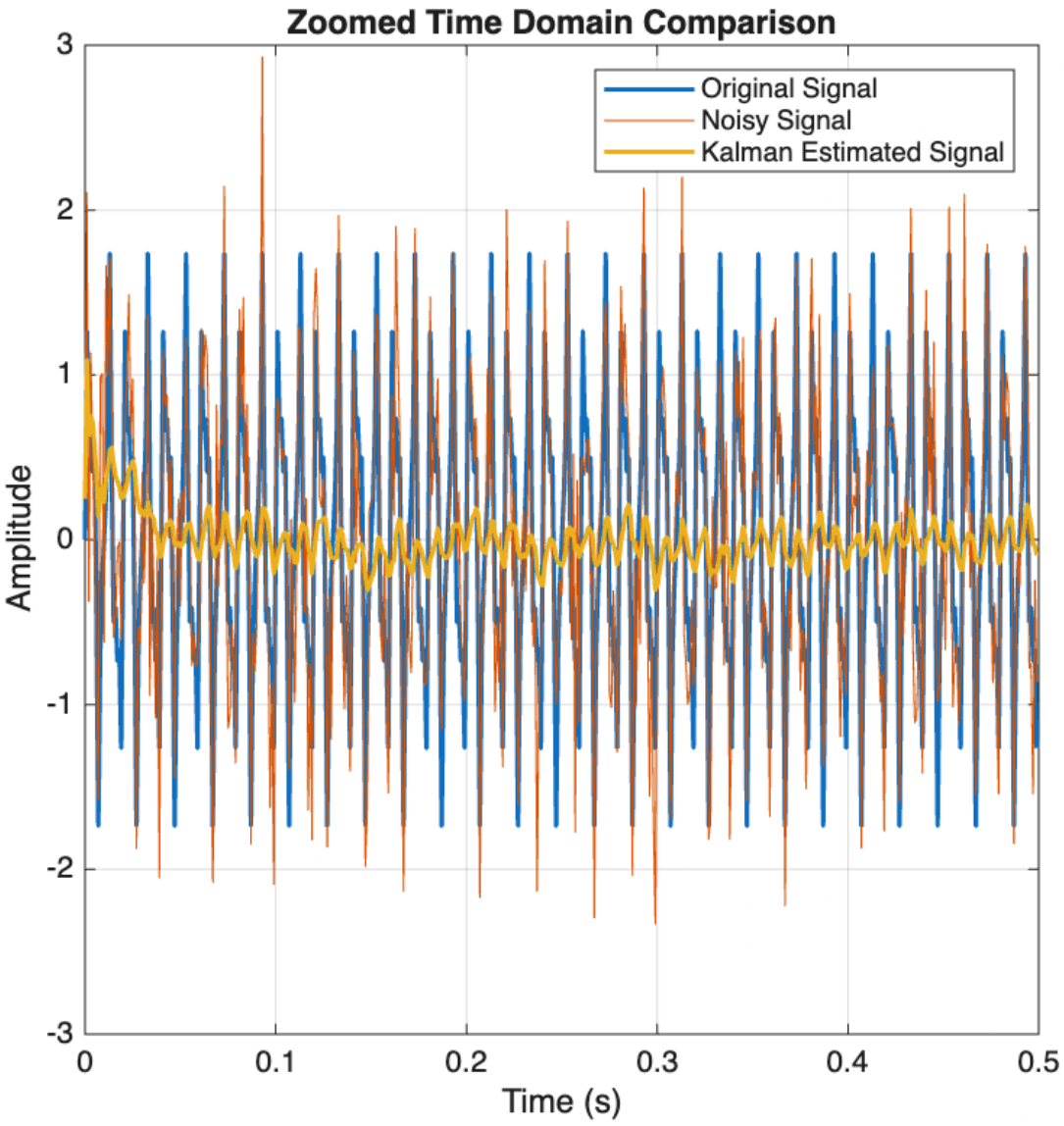
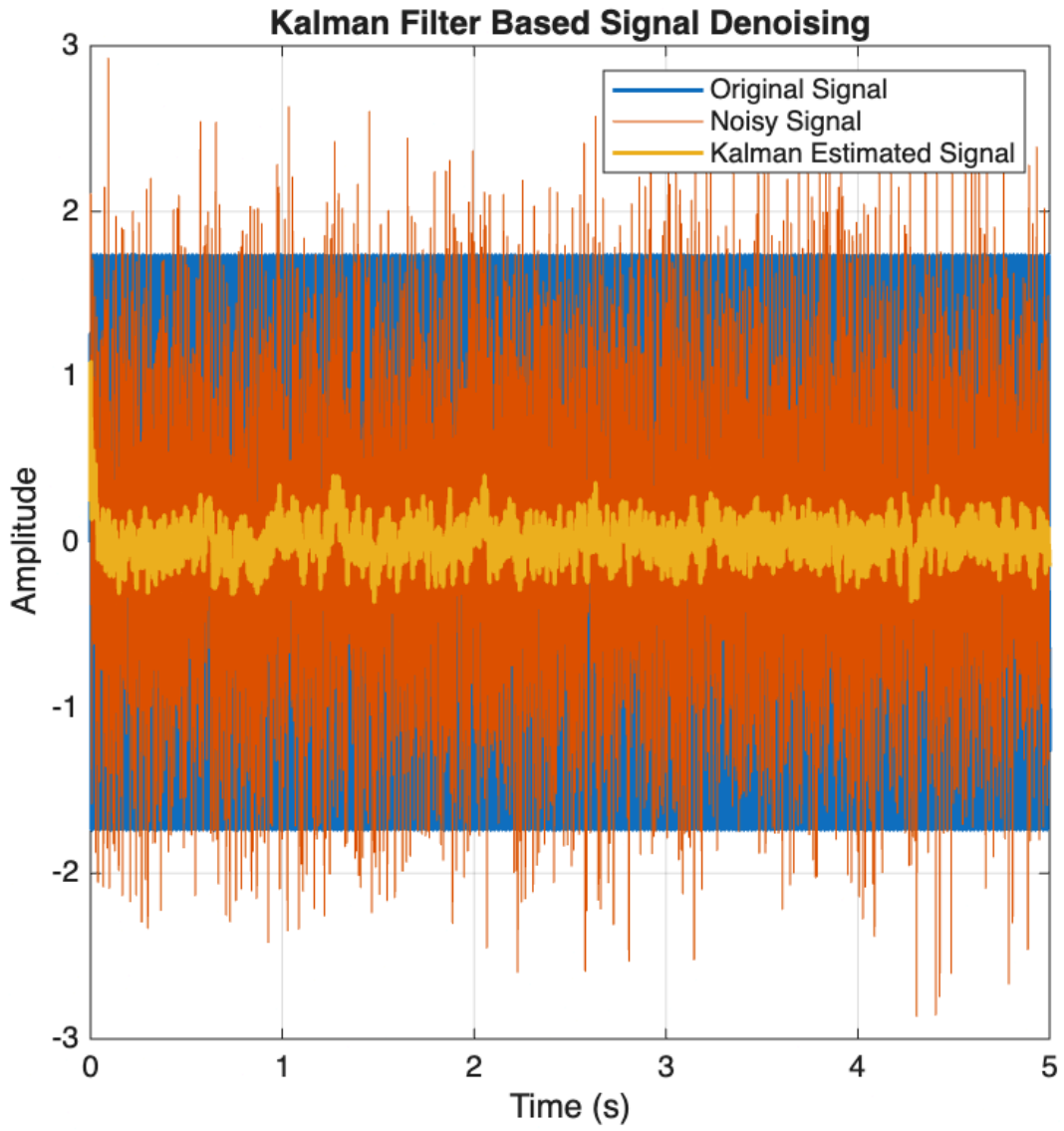

Kalman State Estimate
Estimated Rate of Change
Time (s)
0.4
0.3
0.2
0.1
0
-0.1
-0.2
-0.3
-0.4
-0.5
-0.6
0
1
2
3
4
5


Frequency Spectrum
Original Signal
Noisy Signal
Kalman Estimated Signal
Magnitude
Frequency (Hz)
0
0.05
0.1
0.15
0.2
0.25
0.3
0.35
0.4
0.45
0.5
0
100
200
300
400
500


Zoomed Time Domain Comparison
Original Signal
Noisy Signal
Kalman Estimated Signal
Amplitude
Time (s)
3
2
1
0
-1
-2
-3
0
0.1
0.2
0.3
0.4
0.5


Kalman Filter Based Signal Denoising
Original Signal
Noisy Signal
Kalman Estimated Signal
Amplitude
Time (s)
3
2
1
0
-1
-2
-3
0
1
2
3
4
5